

3) Auto Power Control (APC) Circuit

The APC circuit is difficult to check due to its circuit configuration. This section only describes the theory of its operation.

The system CPU IC-401 yields a parallel 4 bit command signal (see table 17-1) to IC-101 which ultimately controls the APC circuit.

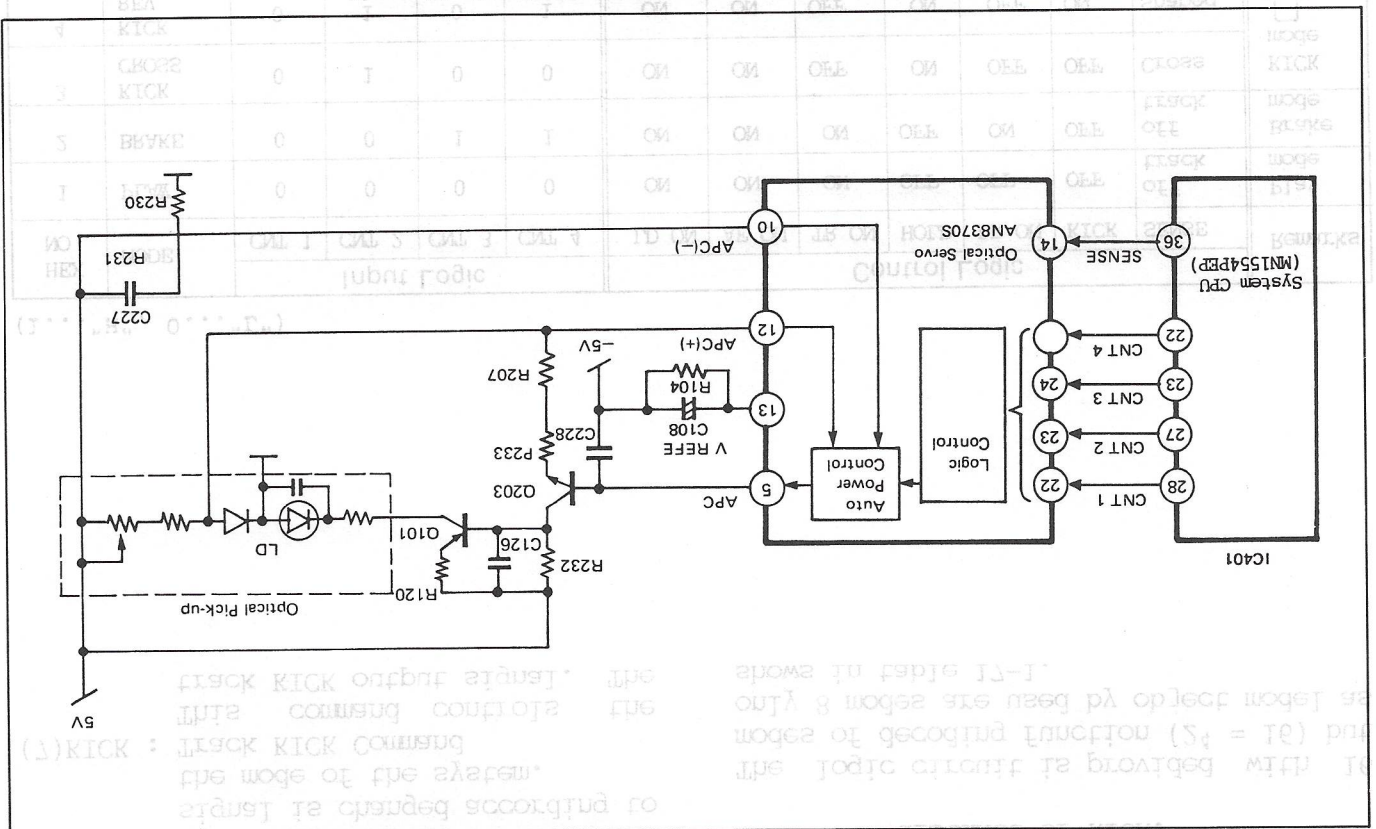
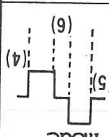
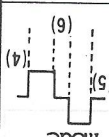


Figure 17-3 Auto Power Control Circuit

When IC-101 receives a Laser On (LD ON) command (H,L,L,L), the logic controller issues a corresponding command to the internal APC block. As a result, pin 5 of IC-101 is set High, which turns Q203, and Q101 "on" and hence activates the LD (Laser diode)

pin 10 of IC-101 is raised (the voltage at the pin 12 of IC-101 is the reference voltage) to diminish the intensity of the laser beam.

Table 17-1 4 Bits Parallel Control Data Functions

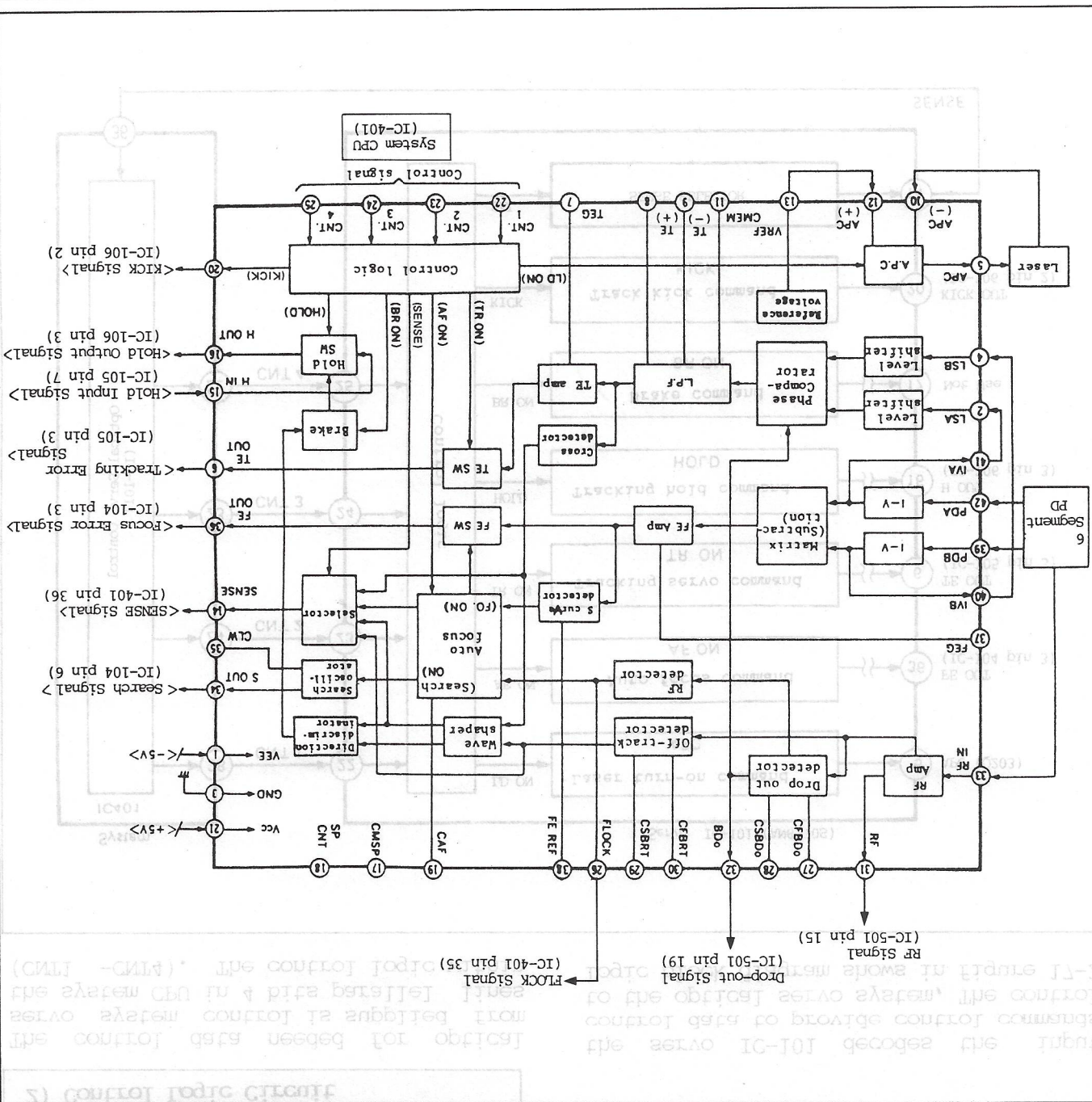
HEX NO	MODE	Input Logic				Control Logic								Remarks
		CNT 1	CNT 2	CNT 3	CNT 4	ID ON	AF ON	TR ON	HOLD	BR ON	KICK	SENSE		
1	PLAY	0	0	0	0	ON	ON	ON	OFF	OFF	OFF	off track	Play mode	
2	BRAKE	0	0	1	1	ON	ON	ON	OFF	ON	OFF	off track	Brake mode	
3	KICK CROSS	0	1	0	0	ON	ON	ON	OFF	ON	OFF	Cross	KICK mode	
4	KICK REV	0	1	0	1	ON	ON	ON	OFF	ON	OFF	Shaped (REV) CROSS	KICK mode	
5	KICK FWD	0	1	1	0	ON	ON	ON	OFF	ON	OFF	Shaped (FWD) CROSS	KICK mode	
6	ACCESS	0	1	1	1	ON	ON	ON	OFF	OFF	OFF	Shaped CROSS	Shaped CROSS	
7	DISC DETECTOR	1	0	0	0	ON	ON	OFF	OFF	OFF	OFF	S-shape curve latch	Disc sense	Forced return
8	STANDBY	1	1	1	1	OFF	OFF	OFF	OFF	OFF	OFF	Shaped CROSS		

(1... "H", 0... "L")

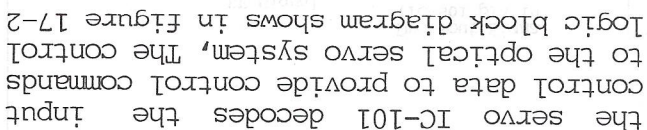
(6)SEL : SENSE Signal Change Command
This command changes the SENSE signal (Confirm the status of the optical servo) transmitted from AN8370S to the system CPU.
The content of the SENSE signal is changed according to the mode of the system.
(7)KICK : Track KICK Command
This command controls the track KICK output signal. The

KICK is classified to KICK FWD (inside to outside) and KICK REV (outside to inside). They are further divided to one track, 10 tracks, and 100 tracks mode according to the distance of KICK.
The logic circuit is provided with 16 modes of decoding function ($2^4 = 16$) but only 8 modes are used by object model as shows in table 17-1.

Figure 17-1 Block Diagram of AN8370S (Optical Servo IC)



The control data needed for optical servo system control is supplied from the system CPU in 4 bits parallel lines (CNT1 - CNT4). The control logic within



(1) LD ON: Laser Ignition Command
This command controls the
turning on and off of the
laser APC circuit.
(2) AF ON: Auto Focus Command
This command controls the
opening and closing of the
focus servo loop, and the on
and off of the search
oscillation switch.

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Figure 16-31 Detection of Tracking Error

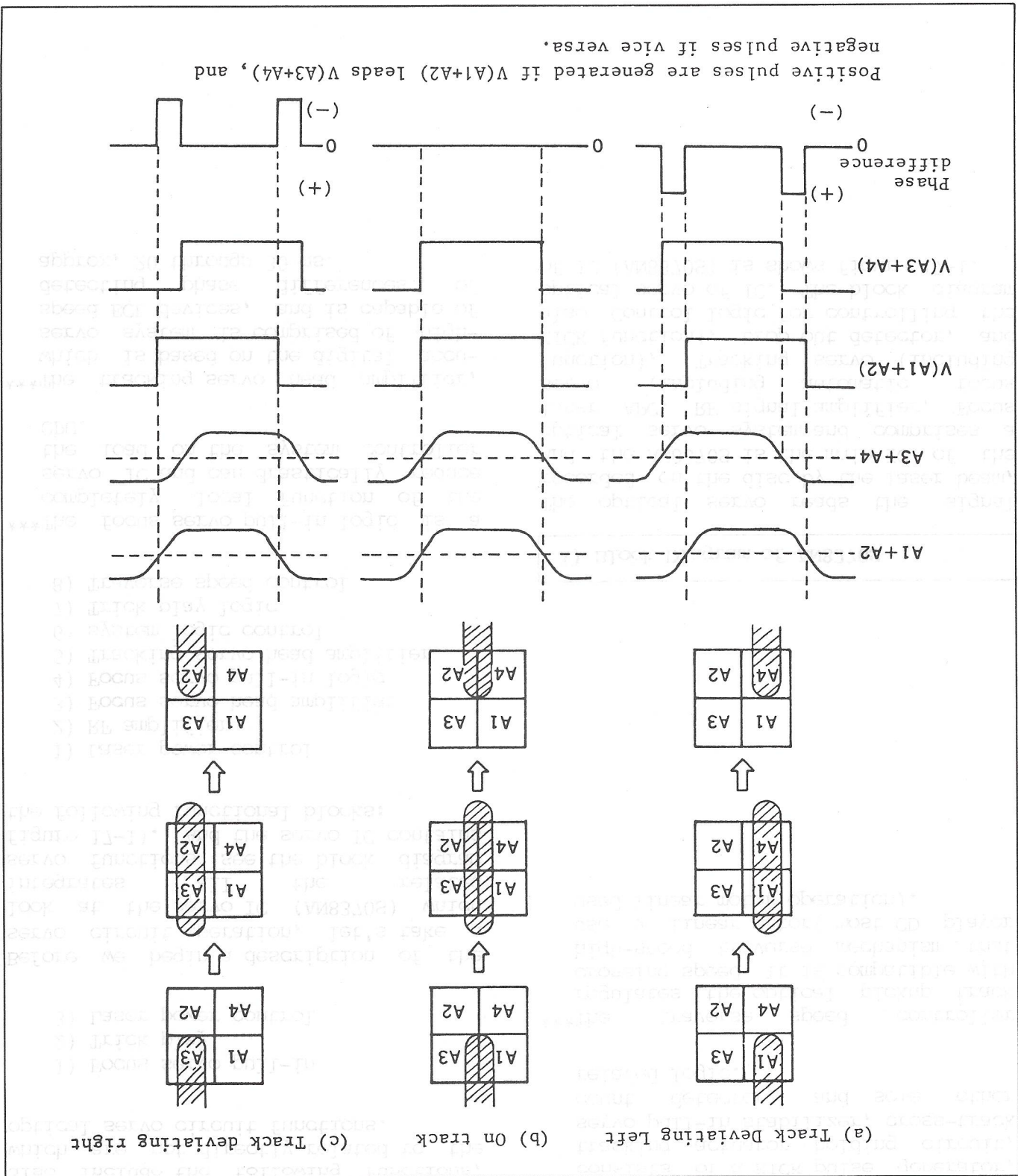
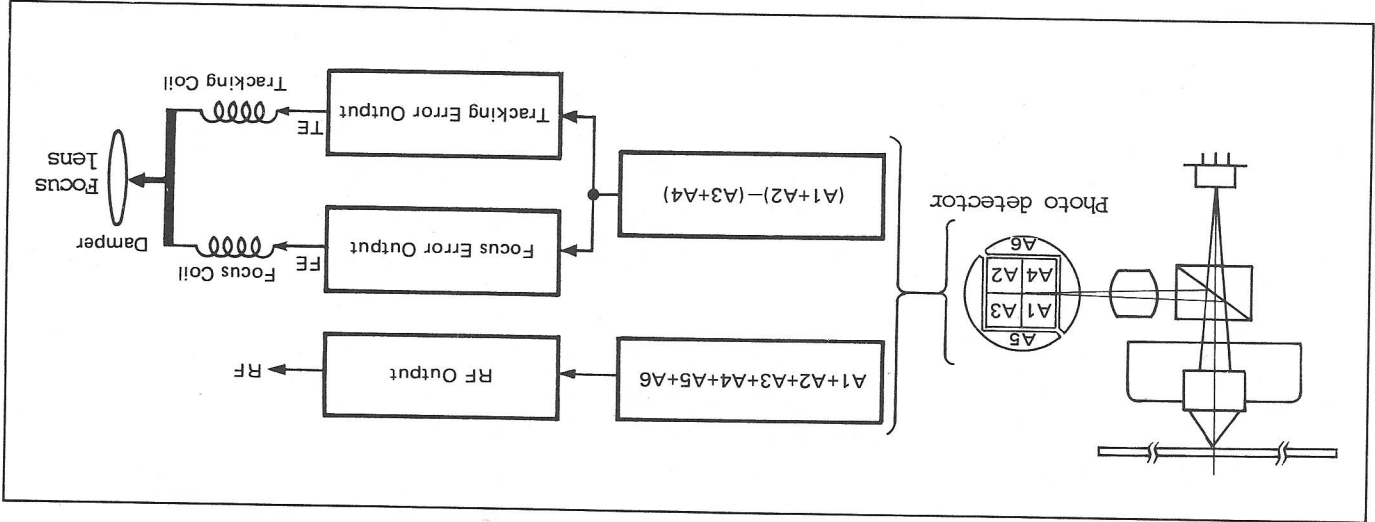
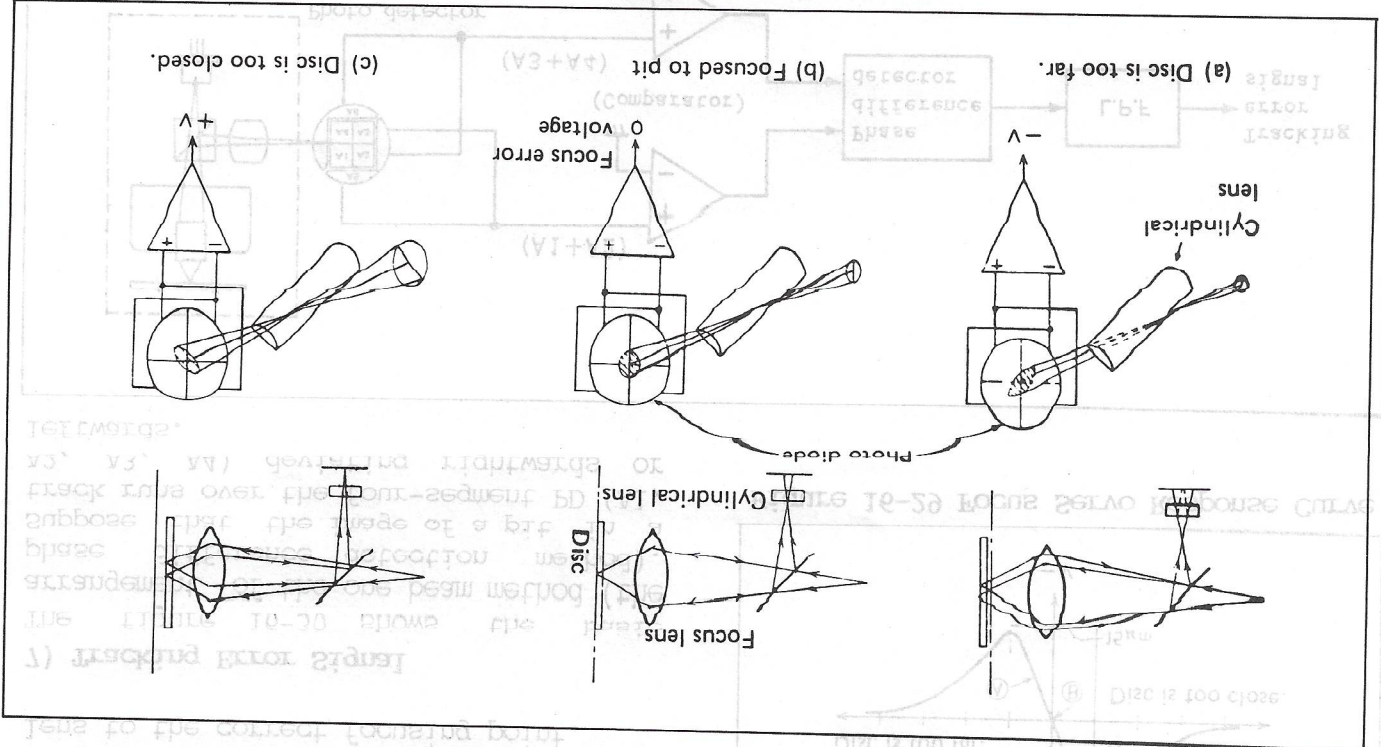


Figure 16-28 Head Amplifier Circuit



They cylindrical lens affects the laser beam in such a way that if it is not focused properly it produces an elliptical shaped spot, the spot falls on a 4 segment photodetector (A1, A2, A3, A4).

Figure 16-27 Focusing Condition



6) Focus Error Signal

The laser must always be focused on the pit area. Since it is impossible to make a disc that has less than 0.11 μm warp

(pit size), the focus lens must move up and down (following the disc) to maintain proper focusing.

FIGURE 16-30 Basic Arrangement

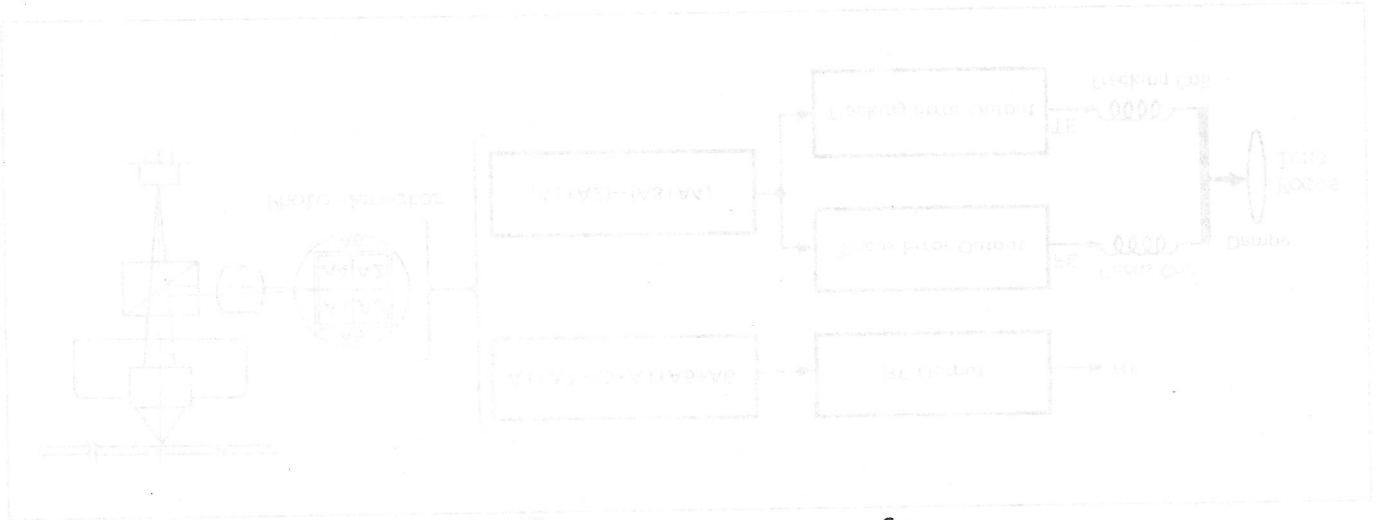


Figure 16-30 Basic Arrangement

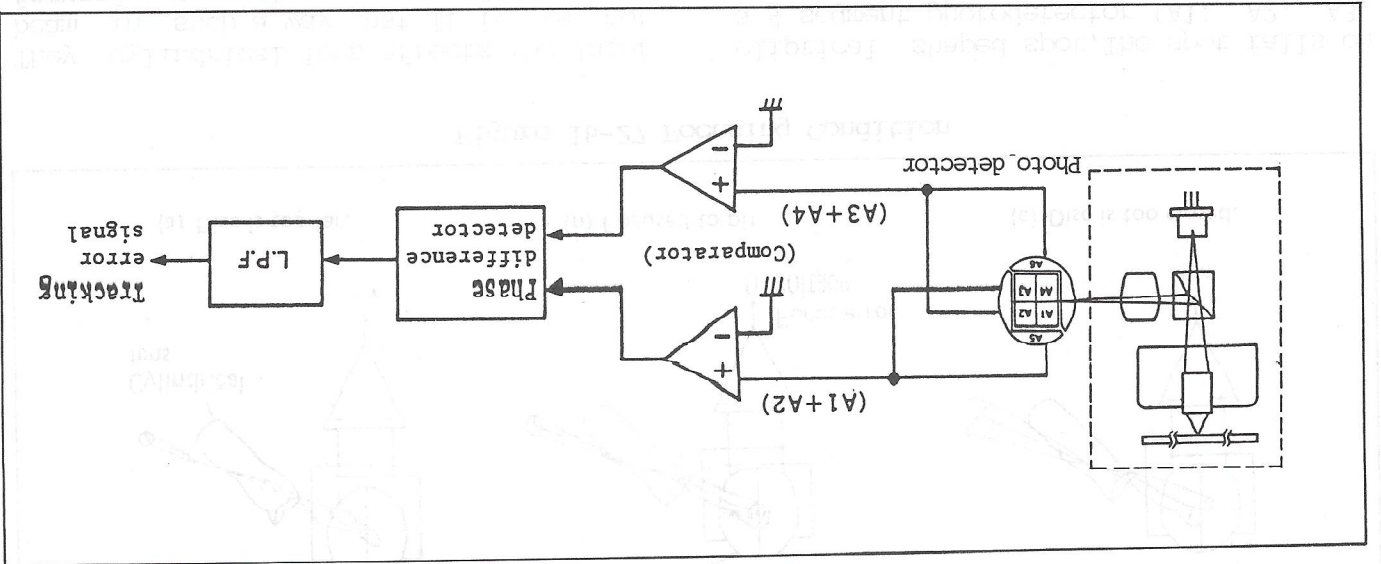


Figure 16-29 Focus Servo Response Curve

The figure 16-30 shows the basic arrangement of the one beam method (the phase difference detection method). Suppose that the image of a pit in a track runs over the four-segment PD (A1, A2, A3, A4) deviating rightwards or leftwards.

7) Tracking Error Signal

The voltage from opposite sections of the detector is added and subtracted from the other sections $(A1 + A2) - (A3 + A4)$. The result is a focus error voltage that is used to drive the focus lens to the correct focusing point.

5) Reading the Disc Information

The figure 16-24 shows a cross sectional view of the disc and laser beam. When the laser beam hits the surface of the disc it is about 1mm in diameter. The transparent layer has a refractive index of 1.5 and used as a lens to reduce the size of the beam to 1 μ m to focus on the pits.

The ratio of the areas of the 1 mm spot on the surface compared to the 1 μ m spot on the pit area is 10,000,000 to 1. Therefore excessive dirt or scratches on the disc surface may affect the reading accuracy.

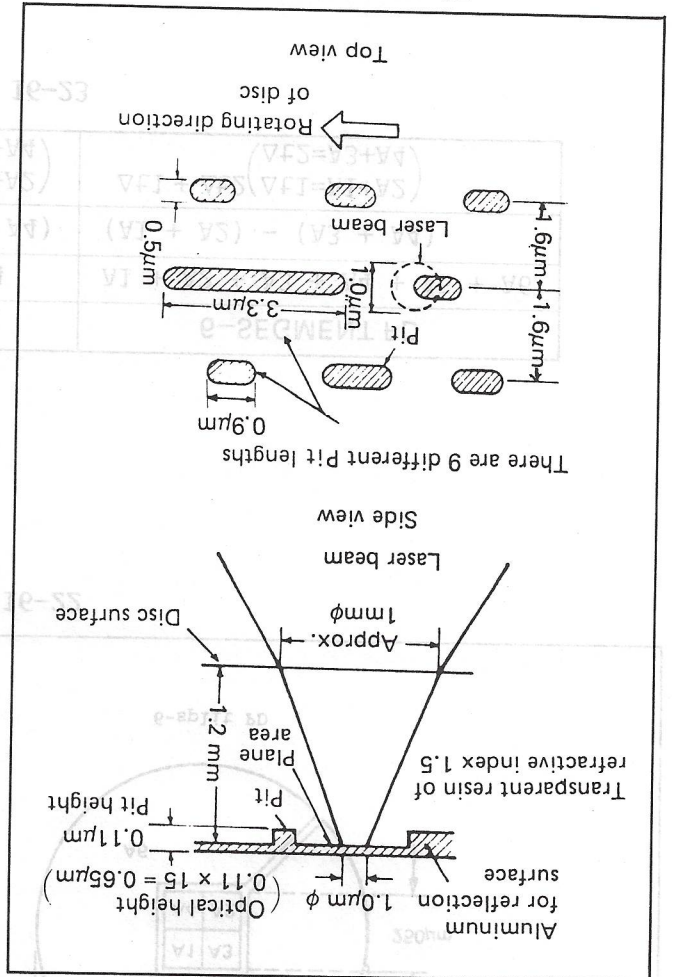


Figure 16-24 Disc

The actual height of a pit is 0.11 μ m. Looking from outside the disc, through the polycarbonate layer (refractive index = 1.5), the perceived height is 165 μ m, this corresponds to about 1/4 wavelength. Therefore, the light is 180 degree out of phase (1/2 wavelength) with the light reflected from the non-pit area as shown in figure 16-25.

Figure 16-25 Pit vs Output Signal

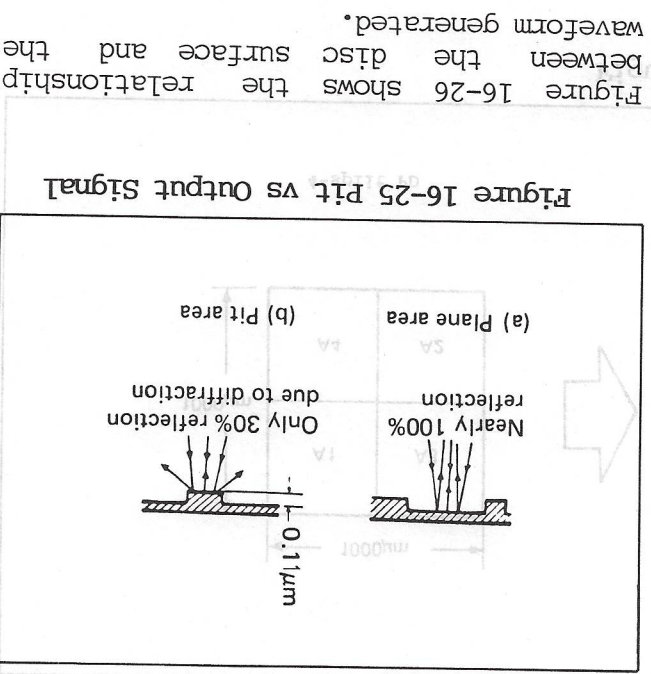
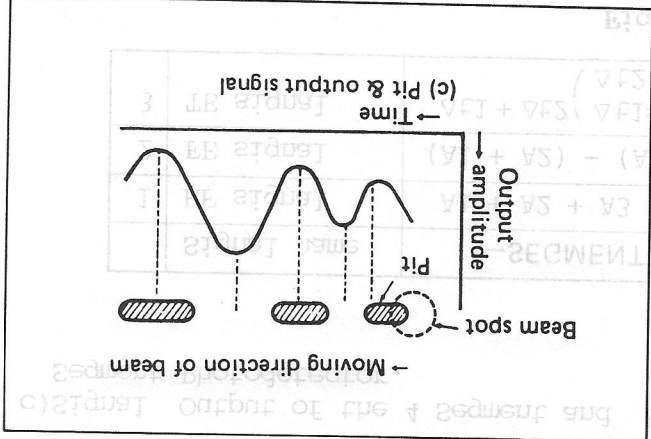


Figure 16-26 Reflection from Disc



When the beam hits the plane (non-pit) area, the light is reflected 100% and returns to the photo detector. This waveform is actually a continuous analog waveform from the digital signal of 4.4218 MHz (bits/sec) recorded on the disc and is called an RF signal.

- CONCLUSION**
1. The PD section for RF signal pickup should preferably have a larger detection surface area.
 2. The PD section for servo signal pickup should preferably have a smaller detection surface area.

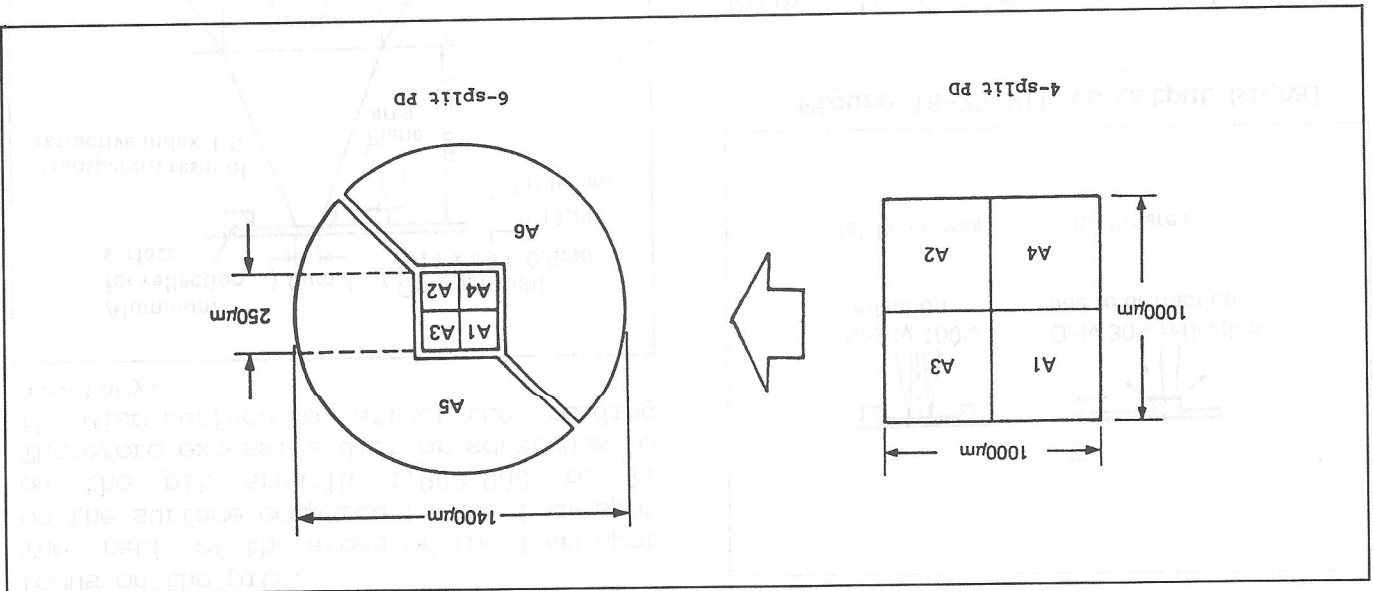
The PD used in the most CD player has achieved the superior RF signal pickup performance of the single beam system for discs with fingerprints, and a stable servo performance comparable to the three beam system for scratched discs.

Figure 16-23

Signal name	4-SEGMENT PD	6-SEGMENT PD
1 RF signal	$A1 + A2 + A3 + A4$	$A1 + A2 + A3 + A4 + A5 + A6$
2 FE signal	$(A1 + A2) - (A3 + A4)$	$(A1 + A2) - (A3 + A4)$
3 TE signal	$\Delta t1 + \Delta t2$ $(\Delta t1 = A1 + A2)$ $(\Delta t2 = A3 + A4)$	$\Delta t1 + \Delta t2$ $(\Delta t1 = A1 + A2)$ $(\Delta t2 = A3 + A4)$

c) Signal Output of the 4 Segment and 6 Segment Photodetector.

Figure 16-22



The above section a) and b) suggests that a larger photodetector is preferable for RF signal pickup, whereas a smaller photodetector is preferable for servo signal pickup. In order to satisfy these incompatible requirements, the most CD player has replaced the conventional 4 segment PD

with a 6 segment PD to successfully achieve more stable play performance. The figure 16-22 shows the size comparison between the 4 segment PD and 6 segment PD for the single beam system.

This type of actuator is insensitive to vibration because it is almost free from friction or play. In addition, since the four wires also serve as electrical leads for the moving coil, the probability of wire break failure is minimized. These features in combination ensure a high reliability for the assembly.

3) Focusing Lens

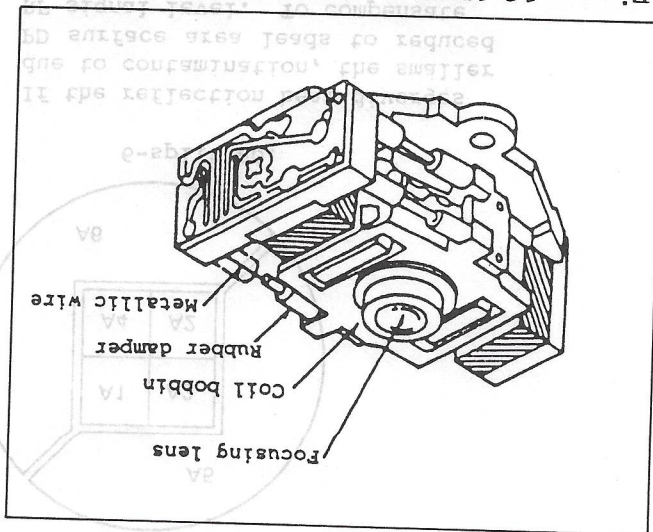
The focusing lens availability include the infinite magnification type for uses in parallel beams, and the finite magnification type for use in divergent beams emitted from a semiconductor laser.

a) Infinite Magnification type: This type has no limitation in the designed optical path length.

b) Finite Magnification type: This type has a fixed distance from the semiconductor laser to the disc surface, and hence requires a lens that matches the construction of the individual optical pickup in which it is used. However this type of focusing lens requires no collimate lens, and renders stable performance due to simple optical system configuration. In addition, the distance from the semiconductor laser to the disc surface can be considerably reduced when compared with the infinite magnification type. However, to compensate for variations of magnification or view angle resulting from focusing lens movement, the finite magnification

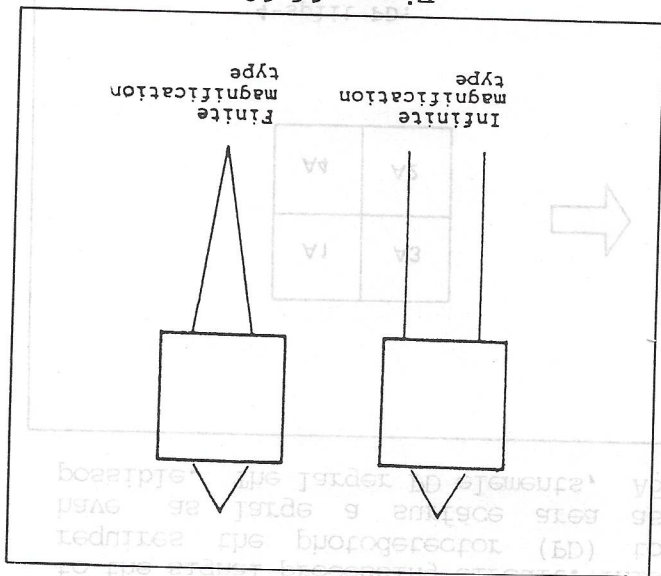
angle resulting from focusing lens variations of magnification or view angle resulting from focusing lens movement, the finite magnification

Figure 16-17 Wire Actuator Construction



This type normally requires stricter specifications than required for the finite magnification type.

Figure 16-18



type normally requires stricter specifications than required for the finite magnification type.

4) Focus and Tracking Error Detection

The optical pickup (photodetector section) yields two basic output signals. One is the musical data signal (RF Signal) which is coupled to the signal processing circuit. The other one is the servo signal used for optical servos. This is coupled to the servo control (focusing and tracking circuit). These signals are affected differently by contamination or scratches on the disc surface, as described below:

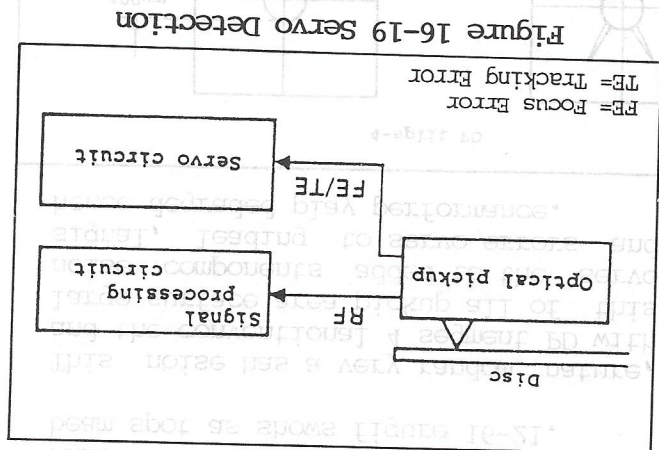
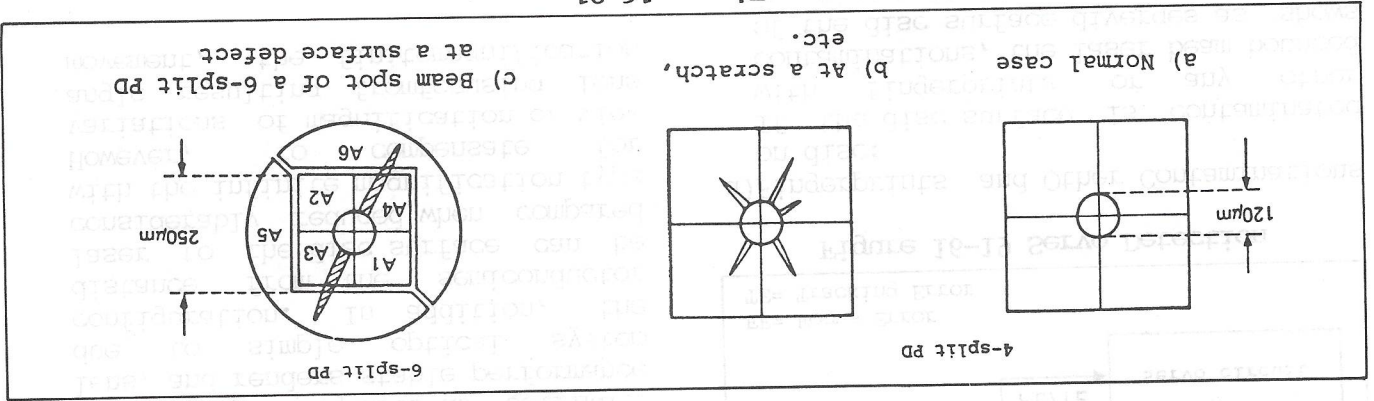


Figure 16-19 Servo Detection

a) Fingerprints and Other Contaminations on disc:

If the disc surface is contaminated with fingerprints or any other contaminations, the laser beam bounced off the disc surface diverges as shown in figure 16-20, this also leads to diverged signal energy.

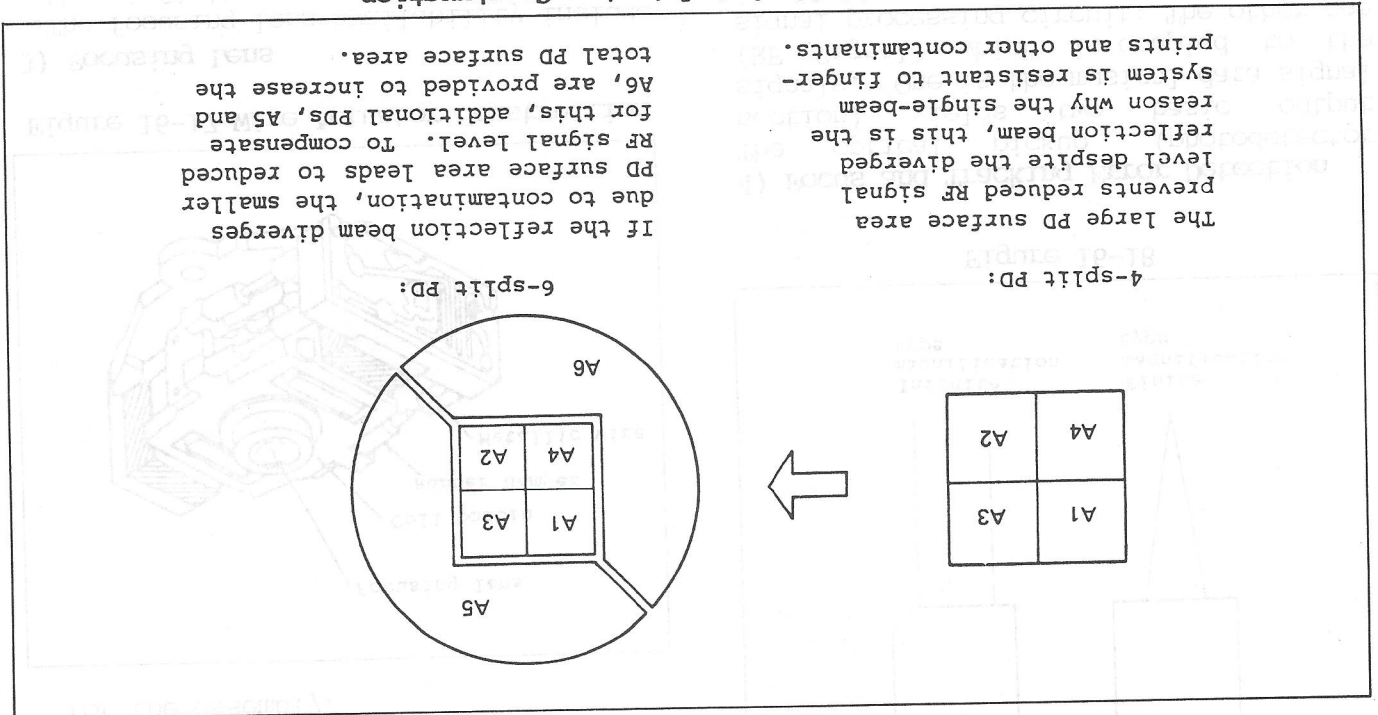
Figure 16-21



(b) Scratches and Foreign Objects

The beam spot diameter on a normal disc surface is approximately 120 μm. If the beam is irradiated at a scratch, or other defects on the disc surface, whisker-like noise occurs around the beam spot as shown in figure 16-21. This noise has a very random nature, and the conventional 4 segment PD with large surface area pickup all of this noise components added to the servo signal, leading to servo errors and hence degraded play performance.

Figure 16-20 Photodetector Construction



The beam, no matter how diverged, must be picked up completely and fed to the signal processing circuit. This requires the photodetector (PD) to have as large a surface area as possible. The larger PD elements, A5 and A6, used in the 6-segment PD (as shown in figure 16-20) pick up the signal energy which has not been picked up by the inner 4 segment PD elements, thus serving to prevent reduced RF signal level.

The following optical components in

particular require careful design:

- Single Beam System
 - 2-Dimension Actuator
 - Focusing Lens
 - Focus and Tracking Error Detection
- The detail of each sections of these components as follows:

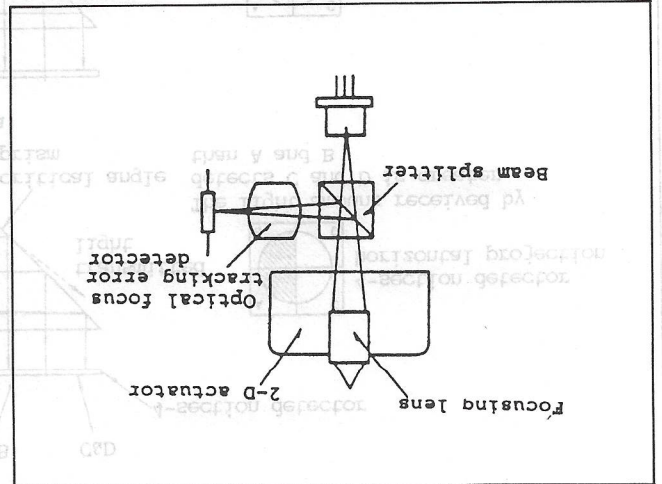


Figure 16-15 Optical Pickup Construction

1) Single Beam System
The single beam detection system includes the push-pull, phasedifference, heterodyne, and pits string on the disc as a single track and detects tracking error from variation of the far-field pattern of the diffracted ray, whereas the phase-difference and heterodyne types detect the variation of far-field patterns of the beam deflected from individual pits.

The push pull type requires a simpler optical system than the three beam technique, but is sensitive to the disc's tilt angle or pit depth and the error signal is affected by the focus lens position. For these reasons, it is rarely used for practical application. The phase difference and heterodyne types eliminate the drawbacks found with the push pull technique. While the optical system is similar to that used in the push pull technique (except for a 4-split photodetector), they have different circuit configurations.

The single beam technique requires a servo circuit a little more complex than that required for the three beam system, but depending on servo circuit design provides better playability than the latter.

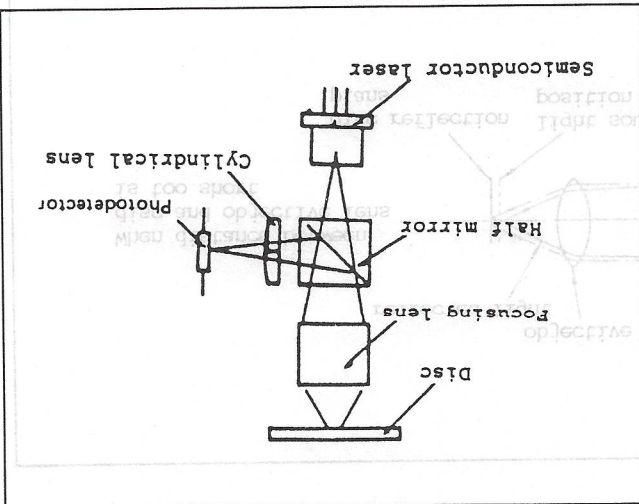


Figure 16-16 Single Beam Construction

2) 2 Dimension Actuator

The 2-D actuator is the only movable component of the optical pickup. It requires precision design because it drives the focusing lens and some other pickup components requiring extreme positional accuracy. Recent 2-D actuators include three types: 4 wire, rotary, and swing arm types. In here will explain the 4 wire actuator as follows:

4 Wire Actuator:

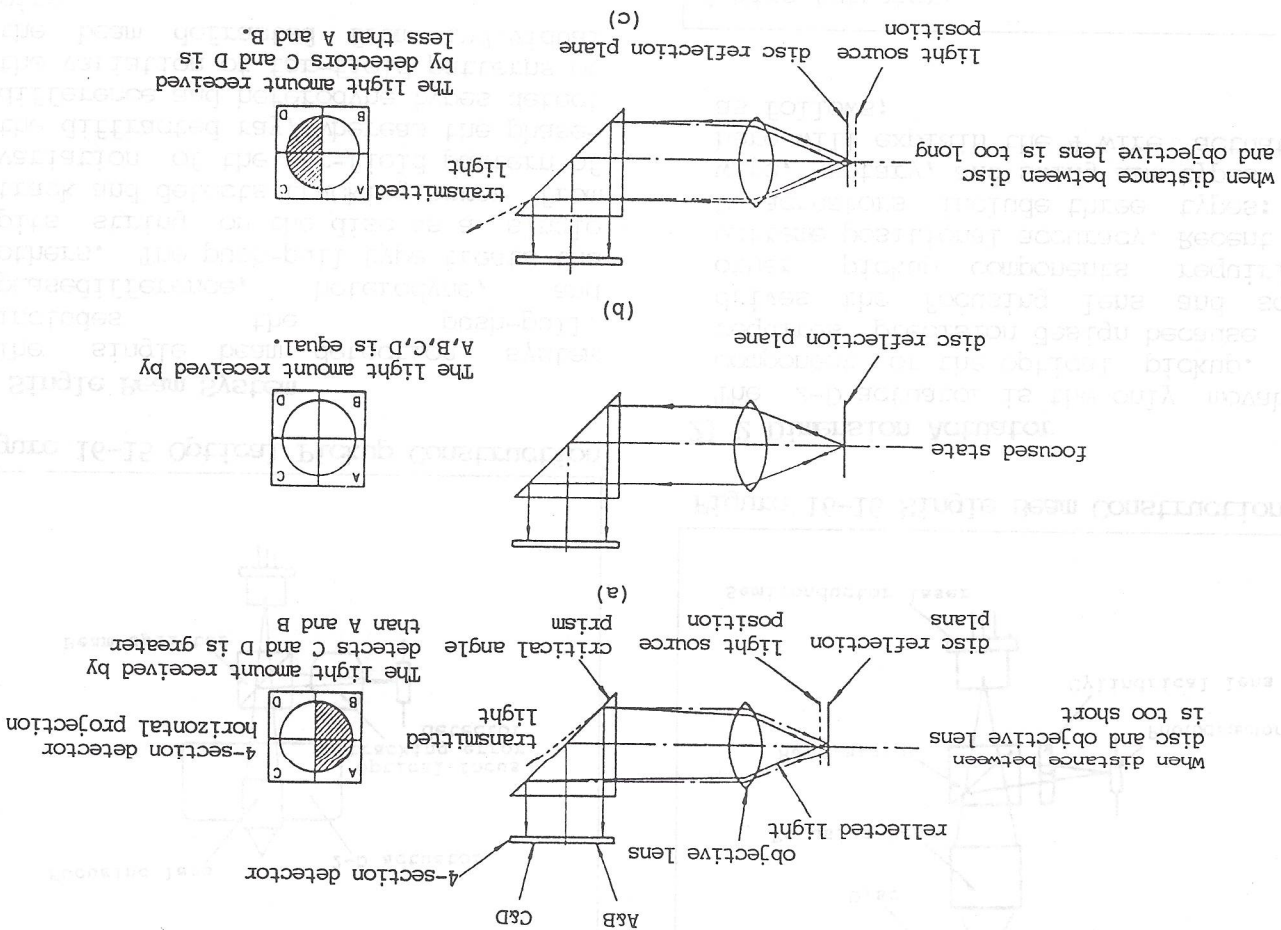
The 4 wire actuator uses four metallic wires to suspend a moving coil bobbin. When the coil is energized, the wires are slack, which lets the coil bobbin move in the focusing and tracking directions. The main point is to drive the bobbin's center of gravity and this gives the four wires and even deformation and keeps the optical axis of the lens straight.

The optical pickup is the heart of the CD player. It pickup modulated laser beam bounced of the disc pits and converts it into electrical data. The pickup (including the traverse mechanism) requires special precision design. On the Technics CD player series, the pickup is integrated onto a single assembly board and is mechanically isolated from the player chassis by a suspension mechanism to prevent floor vibrations and acoustic feedback from the loudspeakers.

As the whole optical system is designed to obtain only P-polarised light, and as the beam is reflected three time, the disc surface position can be detected with extremely high precision.

3. Optical Pickup Constuction and Operation

Figure 16-14 Focus Error Detection



The whole lens system is designed so, that the angle of the incident light in the center of the beam has a critical value. If a parallel beam enters, the incident angle for the whole beam will be the critical angle, so all light will be

reflected. However, when a convergent or divergent light beam enters, only part of the light will be reflected. The critical angle prism is designed so, that the light beam is reflected three time in the prism as shows in figure 16-14.

Glass is the most common optical material at visible and near-infrared wavelengths, but other wavelengths require more exotic materials such as calcium aluminate glass, lithium fluoride etc.

a) The Thin Simple Lens

A thin simple lens refracts an optical ray passing through it as follows:

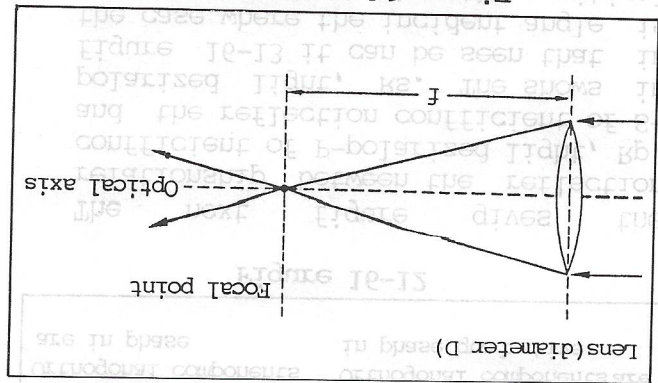


Figure 16-8 Thin Lens

A lens may be either positive (converging) or negative (diverging). The focus point of a lens is that point at which the image of an infinitely distant point source is reproduced.

Both the source and the focus point lie on the lens axis. The focus length (f) is the distance between the lens and the focus point. The f /number of a lens defines its light collecting ability and is given by: Where f is the focus length and D is the diameter of the lens.

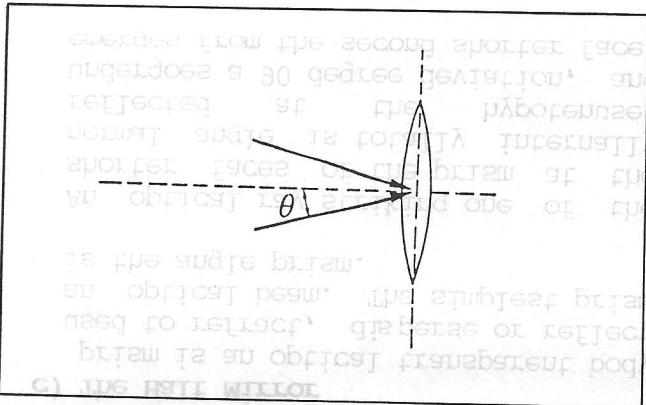
A small f /number denotes a large lens diameter for a specified focus length and a higher light collecting ability of faster "speed" than a large f /number.

Numerical Aperture (NA) is a measure of the acceptance angle of a lens and is given by:

$$NA = n \cdot \sin \theta$$

Where n is the refractive index of the object or image medium and θ is half the maximum acceptance angle.

Figure 16-9



The relation of the focus length (f) to the distances between the lens and the object being imaged (s) and the lens and the focused image (s') is given by the gaussian from of the thin lens equation:

$$1/s + 1/s' = 1/f$$

The combined focus length for the thin lenses in contact or close proximity and having the same optical axis is given by:

$$1/f = 1/f_1 + 1/f_2$$

The relationship between the focus length (f) and the refractive index (n) is :

$$1/f = (n-1) \times (1/r_1 - 1/r_2)$$

Where r_1 is the radius on the left lens surface and r_2 is the radius on the right lens surface.

b) The Cylinder Lens

A cylinder lens is a section of a cylinder and therefore concentrates in only one plane.

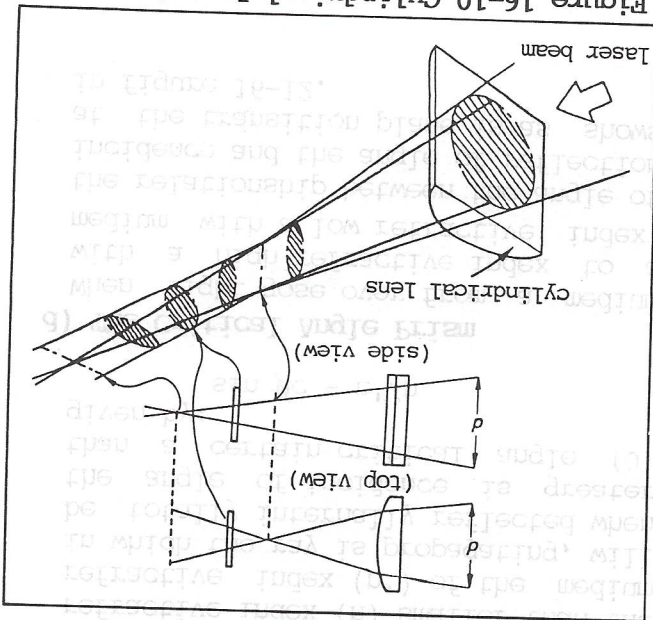


Figure 16-10 Cylindrical Lens Operation

c) The Half Mirror
 prism is an optical transparent body used to refract, disperse or reflect an optical beam. The simplest prism is the angle prism.

An optical ray striking one of the shorter faces of the prism at the normal angle is totally internally reflected at the hypotenuse, and emerges from the second shorter face.

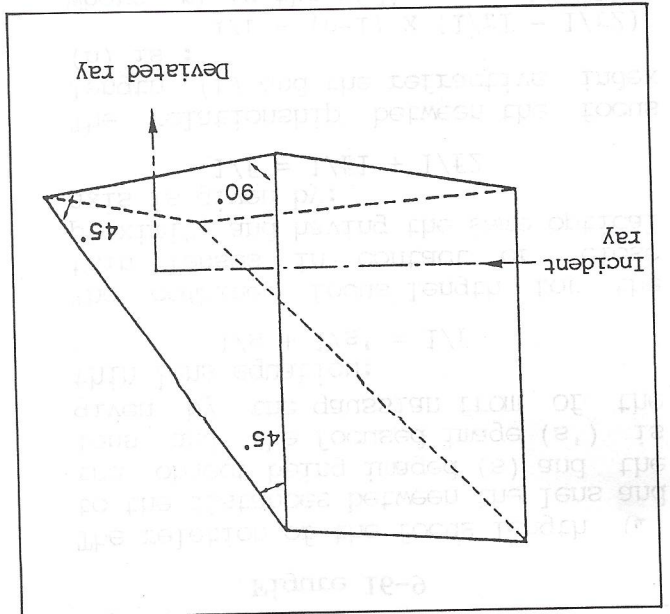


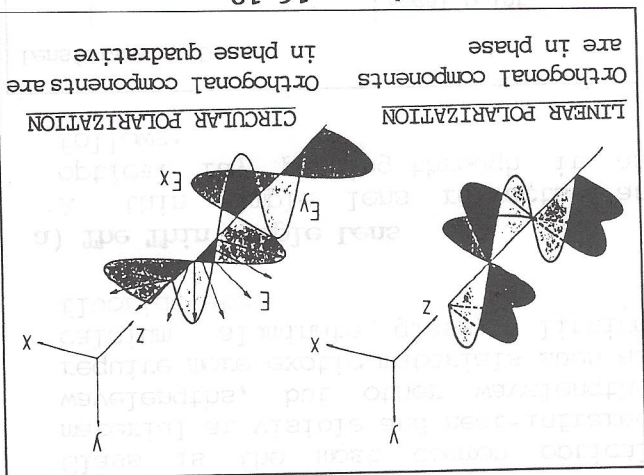
Figure 16-11

Operation of a totally internally reflecting prism is dependent upon the fact that a ray impinging upon the surface of a material having a refractive index (n) smaller than the refractive index (n') of the medium in which the ray is propagating, will be totally internally reflected when the angle of incidence is greater than a certain critical angle (θ_c) given by:

$$\sin \theta_c = n'/n$$

d) The Critical Angle Prism
 When light goes over from a medium with a high refractive index to a medium with a low refractive index, the relationship between the angle of incidence and the angle of reflection at the transition plane is as shown in figure 16-12.

Figure 16-12



The next figure gives the relationship between the reflection coefficient of P-polarized light, R_p , and the reflection coefficient of S-polarized light, R_s . The shows in figure 16-13 it can be seen that in the case where the incident angle is slightly smaller than the critical angle (42 degree), the reflection coefficients will change drastically and there will be an important difference between the reflection coefficients R_p and R_s .

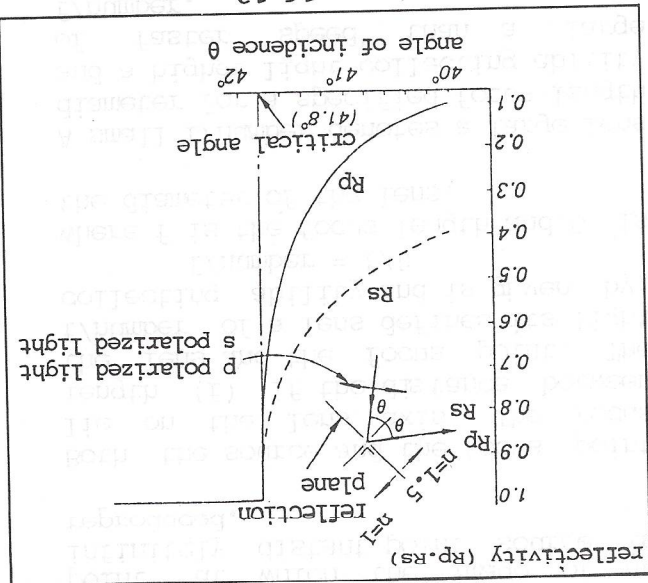


Figure 16-13

This means that, with a small deviation of the incident angle, an important change in reflection can be detected, and an important difference in reflection of P-polarized and S-polarized light. As can be seen in the figure, the change in reflectivity becomes more important for P-polarized light (R_p).

Reflection from a surface may be either diffuse, specular or both. A diffuse reflector has a surface which is rough when compared to the wavelength of the impinging radiation.

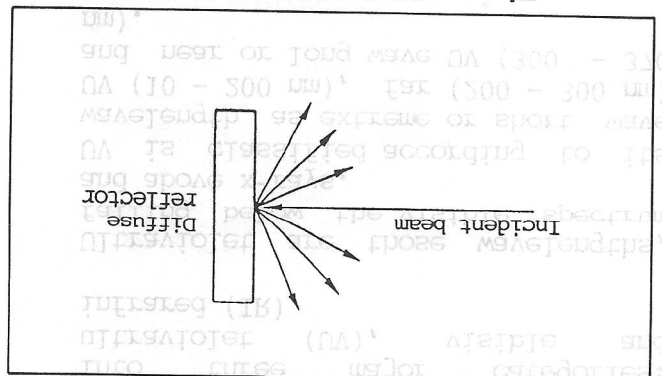


Figure 16-5 Diffuse Reflector

A specular reflection has a surface which is smooth when compared to the wavelength of the impinging radiation. A perfect specular reflector will reflect an incident beam without altering the divergence of the beam. A narrow beam of optical radiation impinging upon a specular reflector obeys two rules:

- (1) The angle of reflection is equal to the angle of incidence.
- (2) The incident ray and the reflected ray lie in the same plane as normal line extending perpendicularly from the surface.

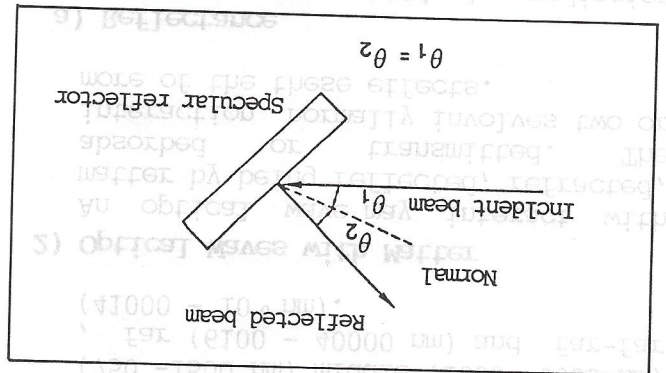


Figure 16-6

b) Absorbance

Some of the optical radiation impinging upon any substance is absorbed by the substance. Absorbance varies according to the properties of the substance and the wavelength and in real circumstances may range from less than 1% to more than 98% ("lampblack" body).

c) Transmittance

Some of the optical radiation impinging upon a substance is transmitted into the substance. The penetration depth may be slight (transmittance = 0) or deep (transmittance more than 75%).

The reflectance (ρ), the absorbance (ϕ), and the transmittance (γ) correspond to:

$$\rho + \phi + \gamma = 1 \text{ (100\%)}$$

d) Refraction

A ray of optical radiation passing from one medium to another is bent at the interface of the two mediums if the angle of incidence is 90 degree. The index of refraction is the sine of the angle of incidence divided by the sine of the angle of refraction. Refractive index varies with wavelength and ranges from 1.0003 to 2.7.

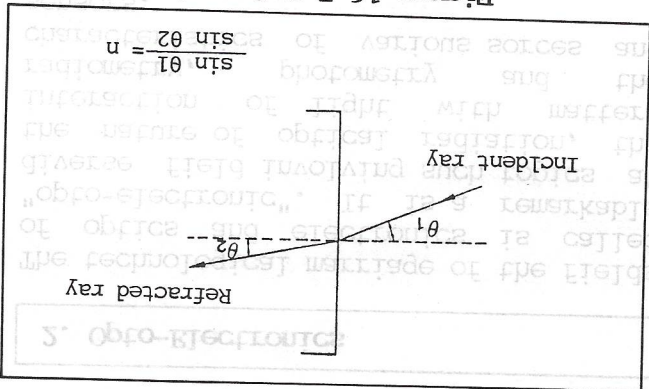


Figure 16-7 Refraction

3) Optical Components

Optical components are used to manipulate and control optical radiation and to provide optical access to various sources and sensors.

Some of the optical radiation impinging upon any surface is reflected away from the surface. reflectance varies according to the properties of the surface and the wavelength and in real circumstances may range from more than 98% to less than 1% (Lampblack body).

a) Reflectance

An optical wave may interact with matter by being reflected, refracted, absorbed or transmitted. The interaction normally involves two or more of these effects.

2) Optical Waves with Matter

Infrared (IR) are those wavelengths above the visible spectrum and below microwaves. IR is classified according to its wavelength as near (750-1500 nm) middle (1600-6000 nm), far (6100 - 40000 nm) and far-far (41000 - 10^6 nm).

The major colour categories are: violet (370 - 455 nm), blue (456 - 492 nm), green (493 - 577 nm), yellow (578 - 597 nm), orange (598 - 622 nm) and red (623 - 750 nm).

Visible are those wavelengths between 370 - 750 nm and they can be perceived by the human eye. Visible light is classified according to the various colours its wavelengths elicit in the mind of a standard observer.

Ultraviolet are those wavelengths, falling below the visible spectrum and above x-rays. UV is classified according to its wavelength as extreme or short wave UV (10 - 200 nm), far (200 - 300 nm) and near or long wave UV (300 - 370 nm).

The optical portion of the electromagnetic spectrum extends from 10 nm (100 Å) to 10^6 nm and is divided into three major categories: ultraviolet (UV), visible and infrared (IR).

Where f is frequency in Hz: c is frequency of light ($=3 \times 10^{10}$ m/sec) λ is wavelength in m

$$f = c/\lambda$$

By convention, electromagnetic radiation is generally specified according to its metric wavelength (λ). The frequency of a specific electromagnetic wavelength can be found from:

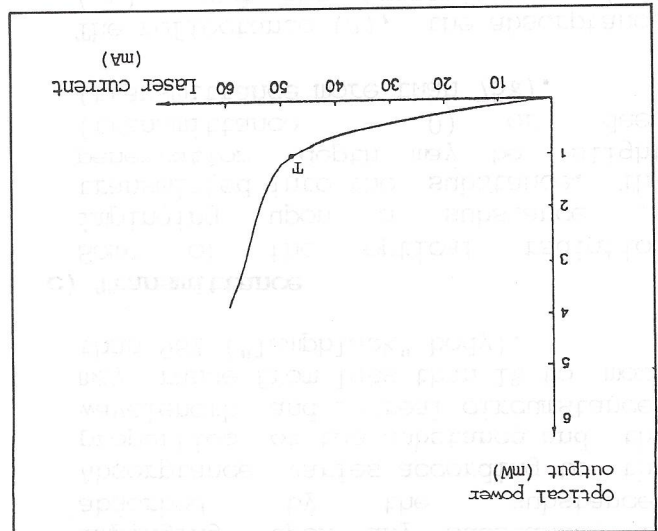
1) The Optical Spectrum

The technological marriage of the fields of optics and electronics is called "opto-electronic". It is a remarkable diverse field involving such topics at the nature of optical radiation, the interaction of light with matter, radiometry, photometry and the characteristics of various sources and sensors.

2. Opto-Electronics

In an (Ga Al) As IID with a double heterostructure (DH), a very narrow PN-junction layer (Ga As) is sandwiched between layers of a different semiconductor (Al Ga As). The properties of this semiconductor confine the electrons and holes to the very narrow junction layer, leading to an inverted population at a low input current. It oscillates at a 780nm wavelength, generating a monochromatic lightbeam.

Figure 16-4



A typical operating curve is shown as follows figure 16-4.